FOUR EQUIVALENT CHARACTERIZATIONS OF LOCALLY FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-finite-type-morphisms; the proof environment immediately following it is its own proof

Editorial note. The label lemma-finite-type-morphisms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the equivalence of the probe and covering characterizations for the finite type property of formal algebraic spaces (lemma-finite-type of the same chapter)
- the Formal Spaces local property criterion for finite type (formal-spaces-lemma-finite-type-local-property)

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at 2574; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ANW: sources/stacks-chow/tags.txt, source lines 13454–13454. Prerequisite pointer invoked at 2598; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04U5: sources/stacks-restricted/pinned-original-source.tex, source lines 2439–2439. Prerequisite pointer invoked at 2596; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally adic* formal algebraic spaces over S. The following are equivalent*

(1) *for every commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

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with U and V affine formal algebraic spaces, $U \to X$ and $V \to Y$ representable by algebraic spaces and étale, the morphism $U \to V$ corresponds to an arrow of WAdm^{adic*} which is adic and topologically of finite type,

(2) *there exists a covering $\{Y_j \to Y\}$ as in Formal Spaces, Definition 0AIM and for each j a covering $\{X_{ji} \to Y_j \times_Y X\}$ as in Formal Spaces, Definition 0AIM such that each $X_{ji} \to Y_j$ corresponds to an arrow of WAdm^{adic*} which is adic and topologically of finite type,*

(3) *there exist a covering $\{X_i \to X\}$ as in Formal Spaces, Definition 0AIM and for each i a factorization $X_i \to Y_i \to Y$ where Y_i is an affine formal algebraic space, $Y_i \to Y$ is representable by algebraic spaces and étale, and $X_i \to Y_i$ corresponds to an arrow of WAdm^{adic*} which is adic and topologically of finite type, and*

(4) *f is locally of finite type.*

Proof. Immediate consequence of the equivalence of (1) and (2) in Lemma 04U5 and Formal Spaces, Lemma 0ANW. □